Joshua L. Bostock

UK: +44 (0)7843 887933
U.S.: +1 206-251-4910
joshua.bostock@outlook.com
github.com/jbostock03
linkedin.com/in/joshuabostock

SUMMARY

Astrophysics Master's student at UCL with a 1st class standing and experience in observational and computational astronomy. Highly adaptable, with strong Python programming skills, statistical analysis, cross-cultural academic experience, and a passion for data-driven problem solving. Seeking to apply my analytical and collaborative strengths in scientific research or data science roles.

EDUCATION

University College London (UCL) — London, UK
MSci Astrophysics — *2022-2026*
- Current mark: 1st (83%)
- Relevant modules: Galaxy Dynamics, Formation & Evolution; Statistical Analysis of Data; Practical Astrophysics & Computing; Advanced Physical Cosmology; Solar Physics; Space Plasma and Magnetospheric Physics; Quantum Physics; Mathematical Methods I, II, & III
- Master's Project: Determining the Period-Luminosity Relation of Gaia Cepheids

University of Washington — Seattle, WA, U.S.
Study Abroad placement as part of competitive university-wide exchange — *2024-2025*
- GPA: 3.82 on 4.0 scale
- Relevant courses: Astronomical Data Analysis; Astrostatistics & Machine Learning; High-Energy Astrophysics; Science of Climate
- Member of the Astronomy Undergraduate Engineering Group, maintaining Manastash Ridge Observatory for outreach events and research
- Coordinated and facilitated discussions with outbound exchange students between universities

The Sixth Form College Farnborough — Farnborough, Hampshire, UK
A-levels – A in Physics, Chemistry, Mathematics & Further Mathematics* — *2020-2022*
- A* in Extended Project Qualification: Understanding the (im)possibility of Dyson Spheres
- Silver in Chemistry Olympiad

The Winston Churchill School — Woking, Surrey, UK
GCSEs – 9 at grade 9 including Mathematics & English Language; 4 at grades 6-8 — *2015-2020*

PROJECTS

Determining the Period-Luminosity Relation of Gaia Cepheids (Master's Project) — *Sept 2025 - Apr 2026*
- Built a full Python pipeline (SciPy, AstroPy, emcee, seaborn) to analyse large astronomical datasets and calibrate regression models for distance estimation, yielding a shift in the Hubble constant by 3.5%, and producing clear visualisations to communicate results to both technical and non-technical audiences
- Implemented non-linear least-squares fitting with full uncertainty propagation and error modelling, improving parameter stability under noisy inputs
- Integrated and cross-matched multi-source datasets (parallaxes, photometric catalogues), applying data cleaning and bias corrections (reddening, selection effects)
- Developed bespoke selection-effect corrections from first principles, accounting for survey incompleteness and detection thresholds
- Applied Bayesian inference using MCMC sampling to validate model robustness; explored ML to benchmark against traditional fitting techniques

Understanding Insolation on Earth — *Apr - May 2025*
- Developed a public-facing educational tool in Jupyter Notebook, improving public understanding of climate systems
- Integrated Matplotlib and ipywidgets for dynamic educational visualisations and intuitive user interaction

Exoplanet Transit Detection — *Apr - Jun 2025*
- Led a student research group in planning and observing exoplanet transits remotely at Apache Point Observatory
- Managed observing strategy and data collection under challenging weather conditions
- Developed full data-reduction including calibration, photometry, and light curve modelling using Python

WORK EXPERIENCE

Johnson Matthey — Online
Virtual Work Experience Programme — *July 2021*
- Participated in workshops on sustainability, innovation, and decision-making in the science and engineering sectors
- Led a cross-functional team to design a net-zero energy house, presenting solutions to a panel of company scientists and engineers
- Gained insight into the role of data-driven analysis in sustainable manufacturing and materials science

Marmalade Game Studio — London, UK
Associate Producer — *July 2019*
- Tested and evaluated gameplay mechanics and user interface designs across multiple platforms
- Logged and categorised bugs and feedback using internal tracking tools, contributing to quality assurance and product refinement
- Joined sprint planning meetings, learning agile development workflows and version control practices

SKILLS

- **Programming & Data Analysis** – Proficient in Python (NumPy, Matplotlib, AstroPy, AstroML, emcee, SciPy, pandas, seaborn), Jupyter Notebooks; DS9; ADQL; experienced in data visualisation, statistical modelling, and image processing
- **Scientific Communication** – Delivered technical presentations at UCL Observatory, reporting on astronomical observations and lab analyses
- **Collaboration & Teamwork** – Contributed to multi-person research projects, such as observing proposals and outreach planning between universities
- **Technical Tools** – Proficient in LaTeX, GitHub integration, GIMP, FITS file handling, Microsoft Office, and Google Workspace
- **Adaptability, Problem Solving, Attention to Detail, Organisation**

HOBBIES & INTERESTS

- **Hiking** – Enjoy exploring new landscapes through day hikes across the UK and abroad, developing resilience and appreciation for the natural world
- **Travel** – Passionate about cultural exchange and adapting to unfamiliar environments, which has enriched my global perspective during academic and personal journeys
- **Photography** – Keen amateur with a focus on astrophotography and long-exposure techniques; proficient in processing RAW and FITS files; featured on UCL's social media channels